LCR damping & resonance

Physics laboratory

Experiment overview

Investigated damped and driven electrical oscillations across different resistor and capacitor combinations; fitted oscillation parameters and compared measured frequencies with theory.

Variables and instrumentation

Resistance, capacitance, voltage response, fitted frequency

Analysis and presentation

Measured-versus-predicted frequency and damping response

Engineering relevance

Shows model validation, error investigation and dynamic-system reasoning.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1BhFvZw6cDRkrks_NFEjAQdYiQ55FqCCZoOufGk_6e0Y/edit